



Customer Retention

Executive Summary

This case study looks at how predictive modeling can be used to improve customer acquisition strategies. The main goal is to identify which customers are most likely to be acquired using the Customer Acquisition and Retention dataset. Being able to make accurate predictions will allow companies to target potential customers, reduce marketing costs, and will allow them to use their resources more efficiently

Multiple classification models were developed and compared including logistic regression, decision trees, and random forests. Logistic regression and decision trees were used as the baseline models to give a starting point for comparing. The random forest was used as the advanced main method because it captures nonlinear relationships and interactions in the variables.

After cleaning and preparing the dataset the analysis focused on important predictors such as industry, revenue, and number of employees. Variables that could cause any data leakage were removed to make sure that the model can work in the real world and give realistic and accurate predictions. We then split the data into training and testing sets to keep the class distribution consistent

The random forest was tuned using 10-fold cross validation and the optimal value for the tuning parameter was found to be $mtry = 2$. The model achieved an accuracy of about 0.735 and a kappa value of 0.383 which shows a moderate performance. But when we compared all models, the logistic regression had the highest accuracy with 83.8%, this outperformed the other models.

The results show that customer acquisition can be effectively predicted using simple models. Factors such as company size and revenue are important in determining acquisition. These findings can improve prediction accuracy and can help companies target potential customers, reduce marketing costs, and also improve marketing strategies.

Problem

Customer acquisition and retention is important for having long term business success.

Companies rely on understanding their customer behavior so they can attract new customers but also maintain steady growth. It is important for companies to be able to identify who can be acquired and retained because not all customers will respond to marketing efforts. By analyzing customer characteristics, companies can get a better understanding on what factors influence acquisition and improve targeting potential customers.

The purpose of this study is to be able to predict whether a customer will be acquired based on the variables in the dataset. This is important because accurate predictions allows companies to focus their marketing efforts on individuals who are more likely to become consistent customers and this reduces marketing costs and improves efficiency.

The main objective in this case study is to build a model that can accurately predict customer acquisition and offer insights into customer duration. It also aims to identify which variables has

the greatest impact on outcomes and determine whether a more advanced model can improve prediction accuracy.

Questions

Specifically, the analysis focuses on the following questions:

1. Can we accurately predict customer acquisition using the dataset?
2. Which factors have the strongest influence on whether or not a customer can be acquired?
3. How well can we estimate customer duration/retention?
4. Which model performs better and is the most accurate?

We used three models to answer these questions, Logistic Regression, Decision Tree, and a Random Forest model. The logistic regression was our baseline model that is based on a linear relationship between the predictors and the outcome. A decision tree model was used to identify nonlinear patterns by splitting the data into groups. A random forest model was used to enhance accuracy and to reduce overfitting in the data.

Outline

The next section outlines the method used in this study and the modeling and evaluation techniques applied to predict customer acquisition. This is followed by a data exploration and insights section which highlights key patterns, distributions, and potential issues within the dataset. A brief literature review is then provided to explain the background behind the models used in this analysis. The report then presents the results of each model which includes performance comparisons and key findings. Finally, the report ends with a summary of the main

insights, business implications, and recommendations for improving customer acquisition strategies.

Methods

For the modeling portion, three classification models were used to predict customer acquisition: logistic regression, decision tree, and random forest. The response variable was acquisition, which was treated as a binary outcome. To avoid data leakage, the final modeling dataset only included variables that would realistically be known before acquisition: industry, revenue, and employees.

The dataset was split into training and testing sets using an 80/20 split with a set seed for consistency. The training set included 401 observations, and the testing set included 99 observations. The class distribution was similar across both sets, which helped make the model comparison more reliable.

Logistic regression was used as a baseline model because it is simple and interpretable. The decision tree was used to capture possible non-linear relationships between predictors and acquisition. Random forest was used as the main model because it combines multiple trees and can reduce overfitting compared to a single decision tree. Model performance was evaluated using accuracy, sensitivity, and specificity.

Data Exploration & Insights

The dataset contains customer-level information that is used to predict if a customer is acquired. We determined the target variable, acquisition, is slightly imbalanced. Approximately 67% of the

customers classified as acquired, whereas, 33% are not acquired. As a result, more customers are acquired than not, this is important to keep in mind when evaluating model performance.

For the exploratory analysis portion, several variables such as frequency, retention, expenditures, profit, and duration were evaluated. These variables demonstrated clear differences between acquired and non-acquired customers. Acquired customers tend to have higher values of the mentioned variables, while non-acquired customers tend to have values close to zero. This indicates that these variables are strongly associated with acquisition.

Numerous of the numeric variables are heavily right-skewed, more specifically the squared variables such as freq_sq and ret_exp_sq. This is expected due to squared variables naturally increasing the spread of larger values. The skewness indicates that a small number of customers have much higher values compared to the rest.

The correlation matrix demonstrated a strong relationship between their corresponding squared terms, such as freq and freq_sq, and ret_exp and ret_exp_sq. This indicates that some predictors may have overlapping information, which can impact how future models utilize these variables.

An important observation from the EDA was that many variables took on zero values for non-acquired customers, while acquired customers have positive values. This suggests that several predictors are highly associated with the outcome and raised significant concerns about potential data leakage, as these variables may reflect behavior after acquisition rather than characteristics beforehand.

Literature Review

Logistic regression is commonly used for binary classification problems because it models the probability of an outcome and provides interpretable results. It estimates how changes in predictor variables affect the likelihood of a customer being acquired. However, it assumes a linear relationship between predictors and the log-odds of the response, which may limit its ability to capture more complex patterns.

Decision trees are nonparametric models that classify observations by splitting the data based on predictor variables. These splits are chosen to best separate the classes, making decision trees useful for capturing nonlinear relationships and interactions between variables. They are also easy to visualize and interpret, which makes them useful for understanding how decisions are made. However, decision trees can overfit the data if they become too complex.

Random forests extend decision trees by building multiple trees and combining their predictions. This typically improves predictive performance and reduces overfitting by averaging across many models. Random forests are able to capture more complex relationships in the data, although they are less interpretable than a single decision tree.

Decision Tree Model

A decision tree model was built using the full dataset to classify whether a customer would be acquired or not. The initial model had perfect accuracy on the test set, with 100% sensitivity and specificity. While this may seem ideal, it is not realistic and suggested potential data leakage issues.

After further investigation, specific variables such as duration, perfectly separated the outcome. All non-acquired customers had a duration of zero, whereas acquired customers had positive

values. This suggests that the model was not properly identifying trends, but instead directly using a variable that already contained the answer.

After removing duration, the model was re-evaluated, but yet still achieved perfect accuracy.

This indicates that other variables such as frequency, cross-buy behavior, and retention-related measures, are also strongly associated with the outcome. These variables suggest that customer behavior after acquisition would not be available at the time of prediction.

Multiple attempts were made to address the issue by removing variables that captured post-acquisition behavior. However, the model continued to show unrealistically high performance. This can be the result of the majority of the dataset containing information closely related to the outcome.

Finally, to resolve this, a final model was built using only variables that would realistically be available prior to acquisition, more specifically industry, revenue, and employees. These variables represent characteristics of the customer rather than the behaviors that occur post-acquisition.

The revised model achieved 72.7% accuracy on the test set, which is more realistic compared to earlier results. The model performed better at identifying acquired customers than non-acquired customers, which is portrayed by the higher specificity compared to sensitivity. This suggests that while the model is able to capture general trends associated with acquisition, it has more difficulty identifying customers who will not be acquired.

Overall, this process highlights the importance of identifying and addressing data leakage. While the initial models achieved perfect accuracy, they did not provide meaningful insights. The final

model offers a more realistic evaluation of performance and better reflects how the model would perform in a real-world setting.

Results

The logistic regression model had the strongest overall performance, with an accuracy of 83.8%. The random forest model had an accuracy of 75.8%, and the decision tree had an accuracy of 72.7%. Based on accuracy, logistic regression was the best model for predicting customer acquisition in this dataset.

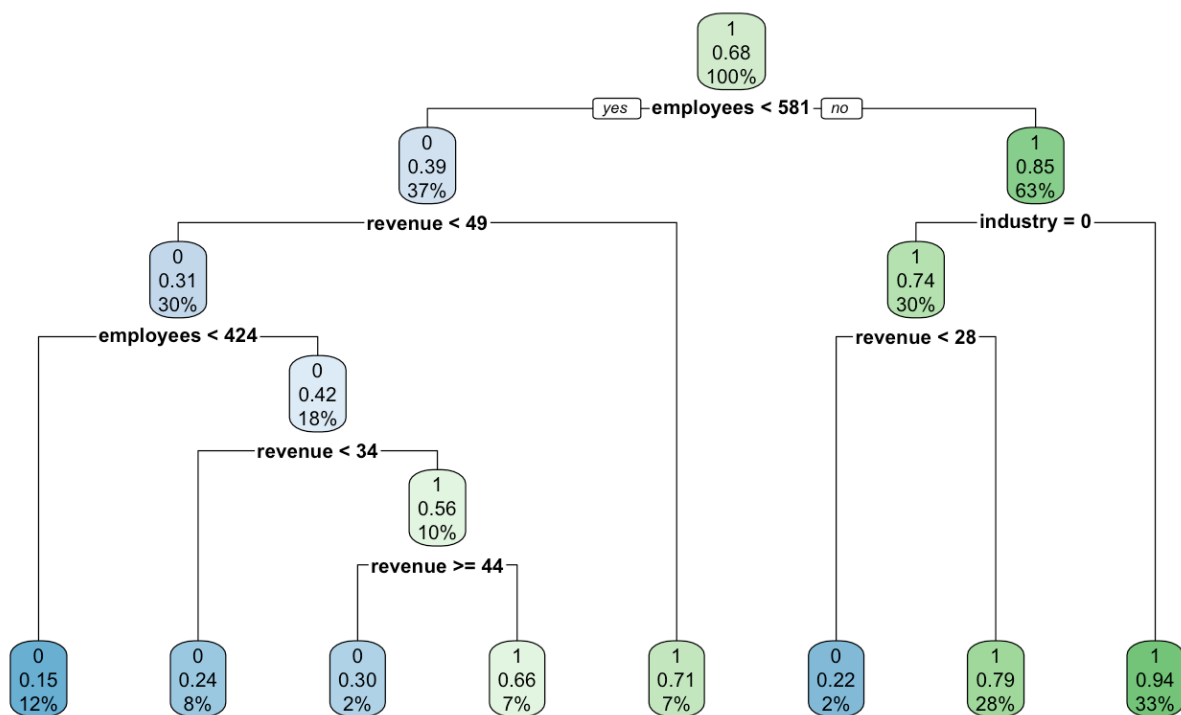
Table 1. Model Performance Comparison

Model <chr>	Accuracy <dbl>	Sensitivity <dbl>	Specificity <dbl>
Logistic Regression	0.8383838	0.65625	0.9253731
Decision Tree	0.7272727	0.46875	0.8507463
Random Forest	0.7575758	0.62500	0.8208955

As shown in Table 1, logistic regression achieved the highest accuracy, followed by random forest and decision tree. While the random forest performed better than the decision tree, it did not outperform logistic regression.

The decision tree model provides a visual representation of how the predictors are used to classify outcomes. As shown in Figure 1, variables such as employees and revenue are used to split the data, indicating that company size and financial characteristics play an important role in predicting acquisition.

Figure 1. Decision Tree for Customer Acquisition



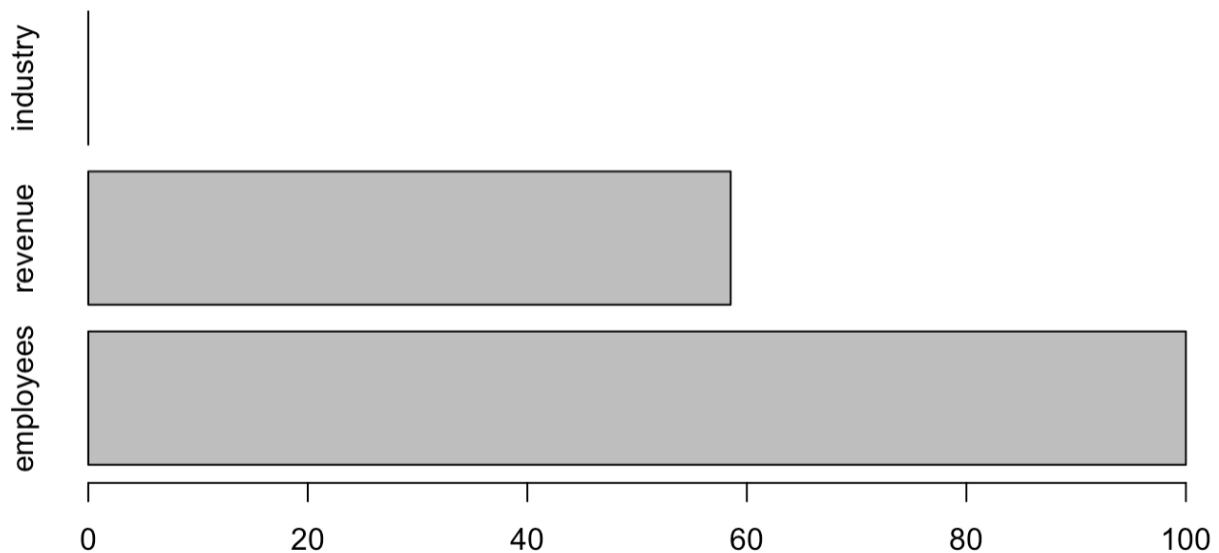
The tree structure suggests that firms with higher employee counts and revenue are more likely to be classified as acquired customers.

The random forest model was tuned using cross-validation. The best value for mtry was 2, with a cross-validated accuracy of about 73.5% and a Kappa value of 0.383. On the test set, the random forest accuracy improved slightly to 75.8%, suggesting that the model generalizes reasonably well to new data.

The variable importance results from the random forest showed that employees was the strongest predictor, followed by revenue, while industry had very little importance in the model.

Figure 2. Random Forest Variable Importance

Random Forest Variable Importance



This suggests that company size and financial scale are more important for predicting acquisition than the specific industry category.

Overall, the results suggest that the relationship between the predictors and acquisition is relatively straightforward. Larger firms with higher revenue appear more likely to be acquired, indicating a clear relationship between firm size and acquisition probability. This helps explain why logistic regression performed better than the more complex random forest and decision tree models. This suggests that acquisition is largely driven by firm scale, where larger and higher-revenue companies are systematically more likely to be acquired than smaller firms.

Conclusions

Ultimately, logistic regression was the best-performing model for predicting customer acquisition, with an accuracy of 83.8%. Although random forest was the main model for this

section, it did not outperform logistic regression. The decision tree had the weakest performance, likely because a single tree can oversimplify the relationships in the data.

The main business insight is that larger companies, especially those with more employees and higher revenue, appear more likely to be acquired. This means the firm could use these variables to help prioritize which potential customers to target.

In future work, the analysis could be extended by testing additional methods such as LASSO regression, gradient boosting, or random forest with more non-leakage predictors. Adding more behavioral or customer engagement variables could also improve prediction accuracy. These results directly support the goal of improving customer targeting by identifying which customers are most likely to be acquired.